

SIMULATION REPORT

Week 2 Final: Multi-Agent Limit Order Book Simulation

CONFIGURATION:

- Total Agents: 100
- Random Seed: 42
- Simulation Time: 30 minutes (1800 seconds)
- Snapshot Interval: 1 second(s)

SCENARIOS:

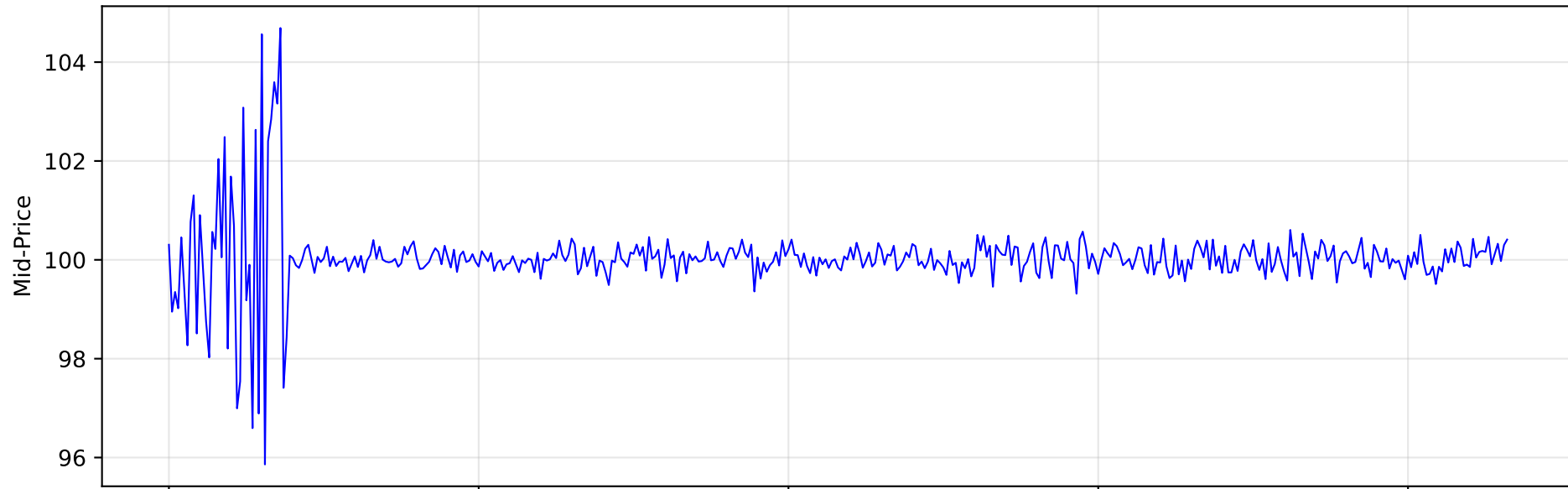
- Scenario A: 100 NoiseAgents
- Scenario B: 80 NoiseAgents + 20 MarketMakerAgents
- Scenario C: 80 NoiseAgents + 20 MomentumAgents

AGENT PARAMETERS:

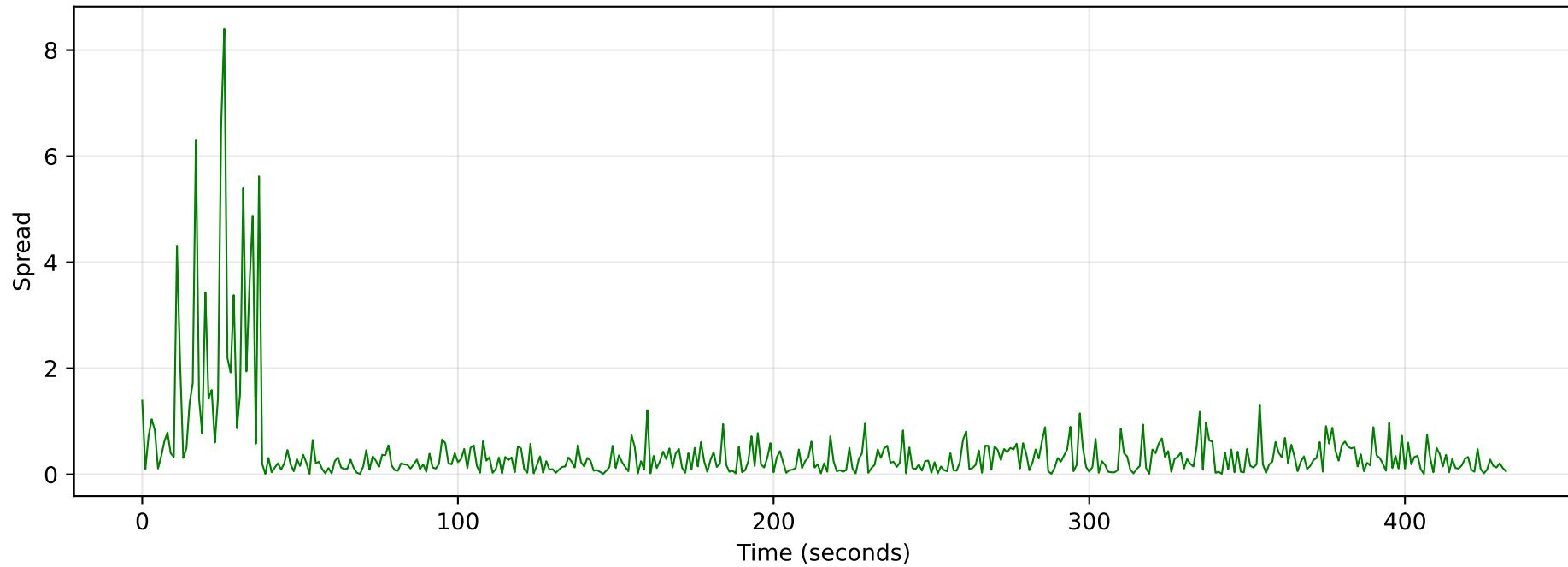
- NoiseAgent: cash=10000, inventory=50, volatility=0.1
 - MarketMakerAgent: cash=10000, inventory=50, spread=2.0
 - MomentumAgent: cash=10000, inventory=50, lookback=50
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Scenario A: 100 NoiseAgents

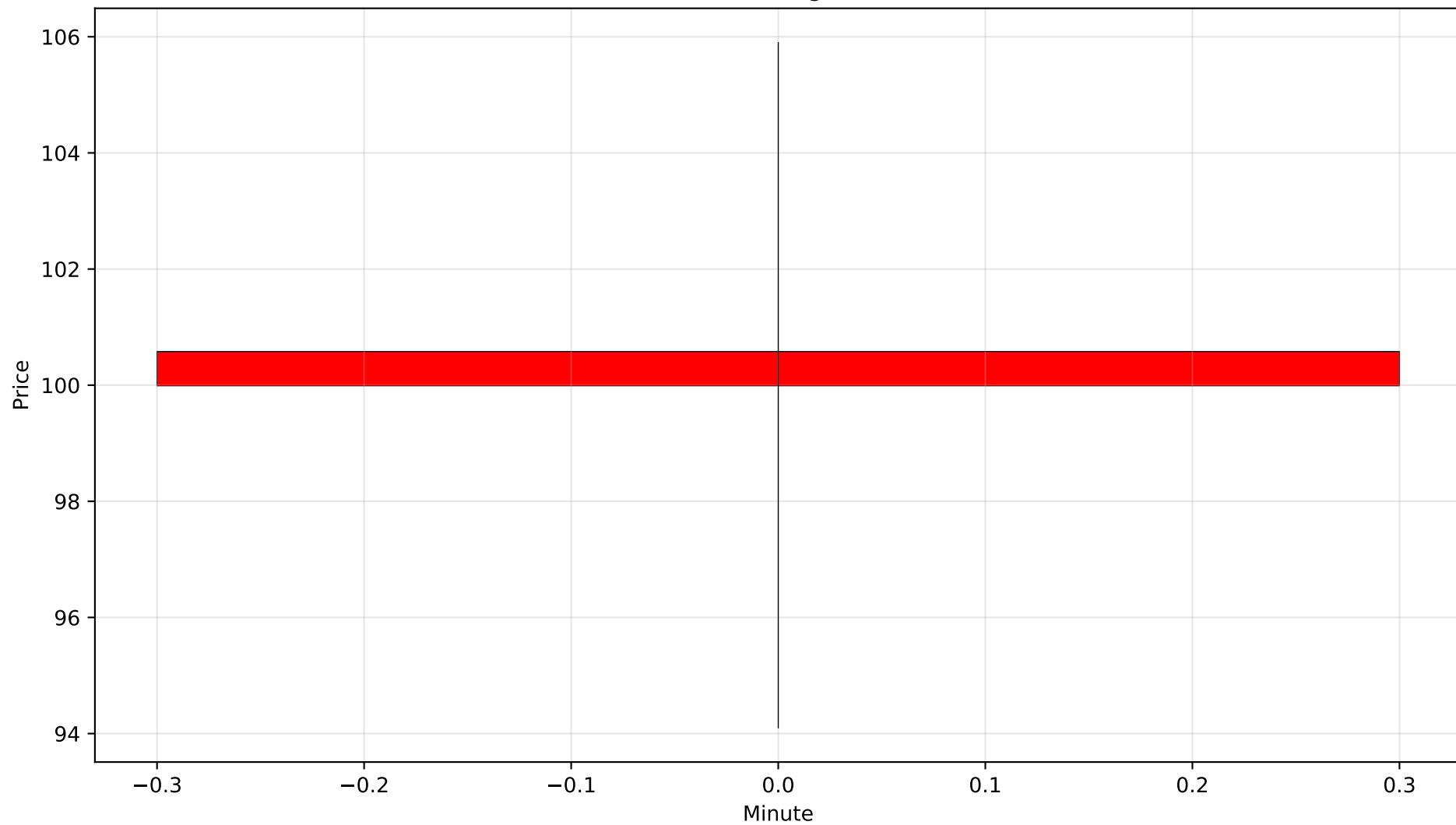
Mid-Price Over Time



Bid-Ask Spread Over Time

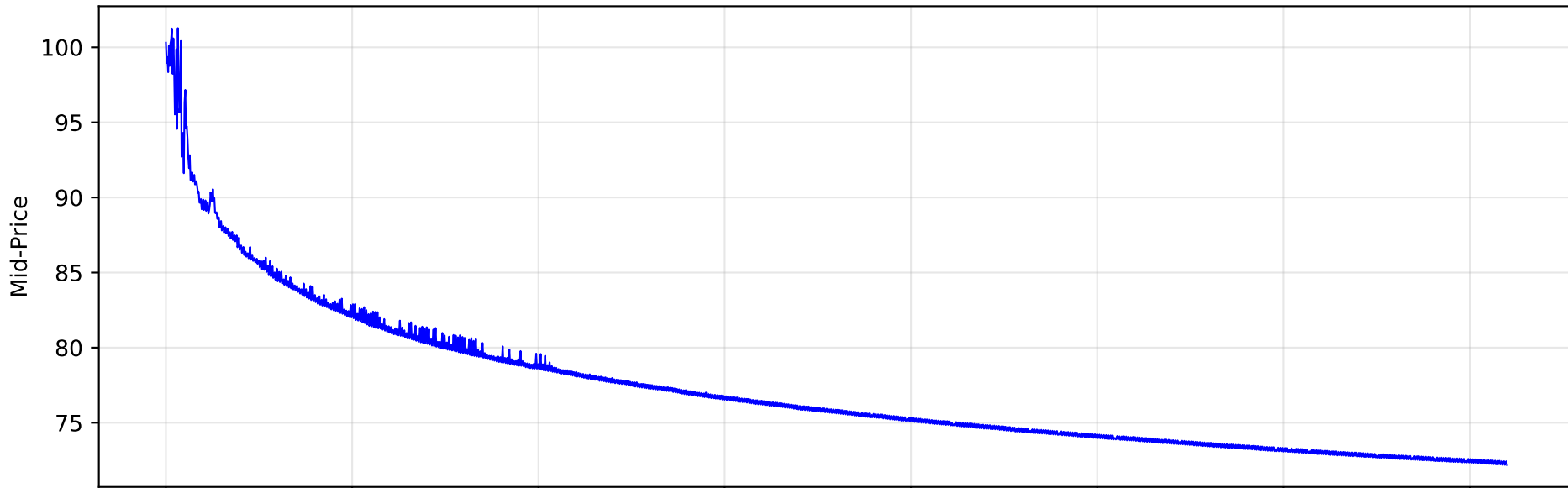


Scenario A: 100 NoiseAgents: 1-Min OHLC

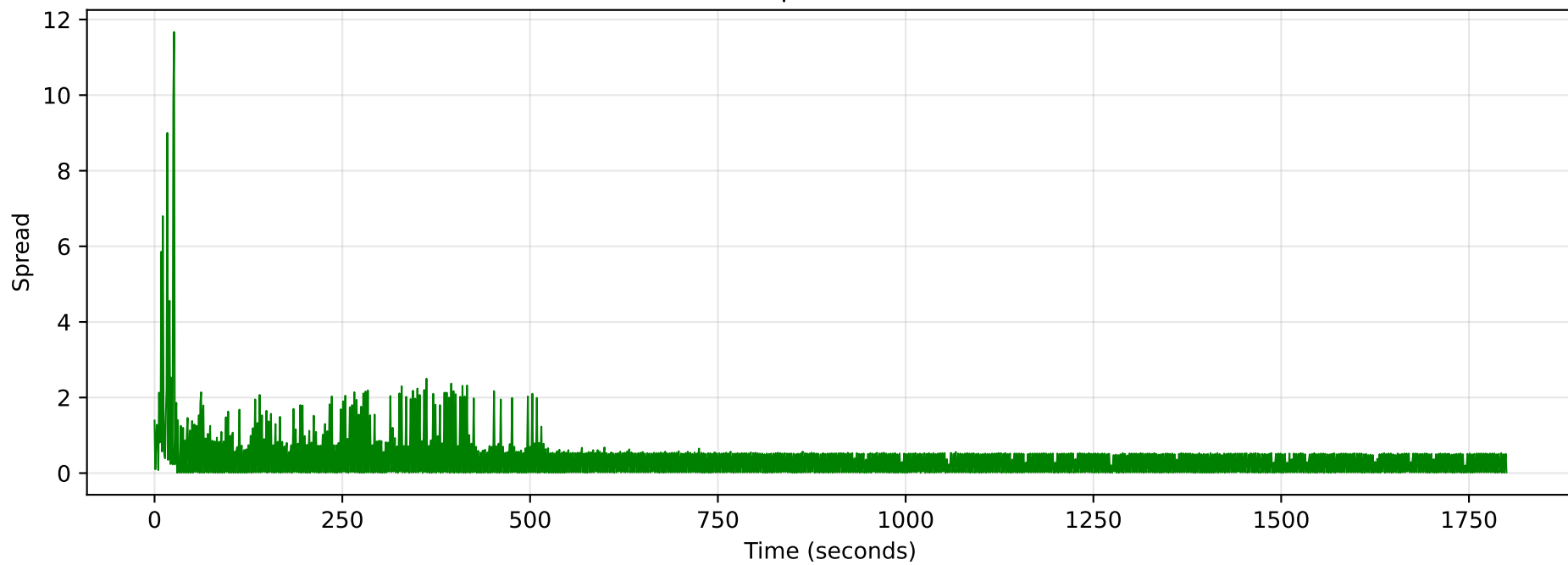


Scenario B: 80 Noise + 20 MM

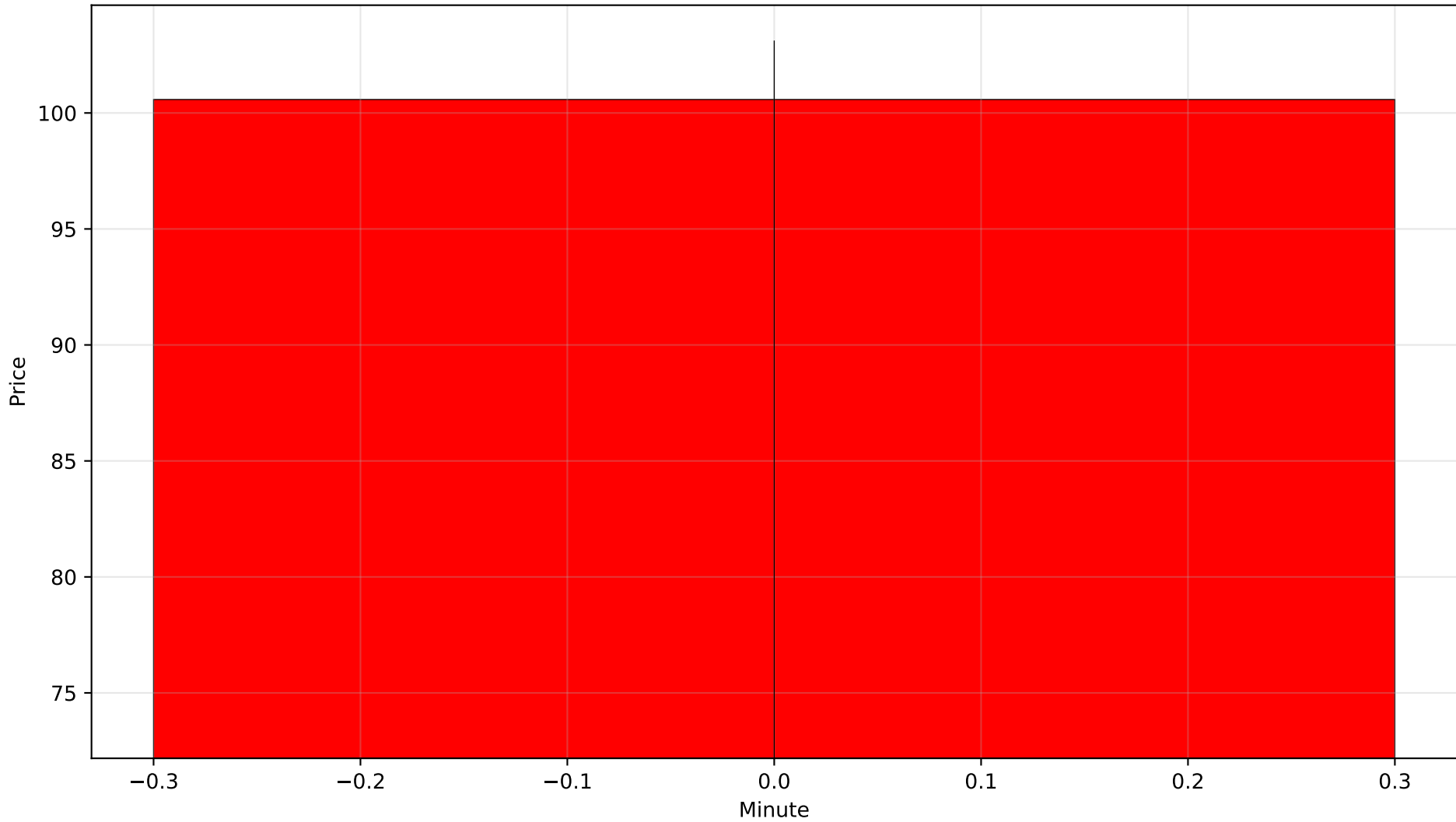
Mid-Price Over Time



Bid-Ask Spread Over Time

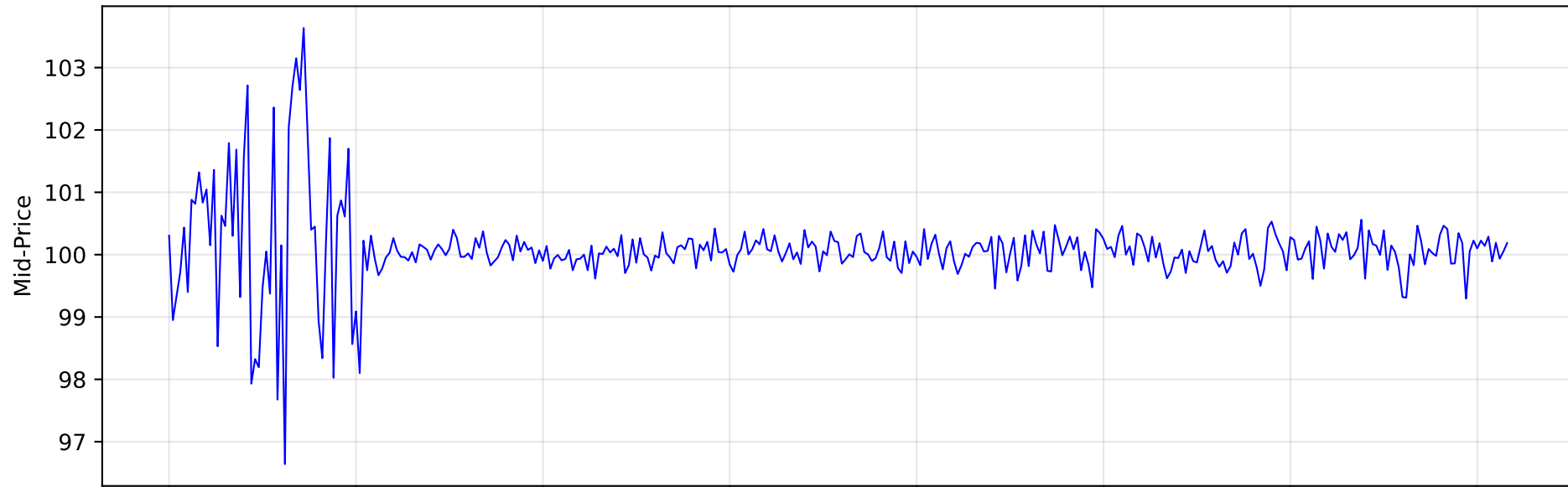


Scenario B: 80 Noise + 20 MM: 1-Min OHLC

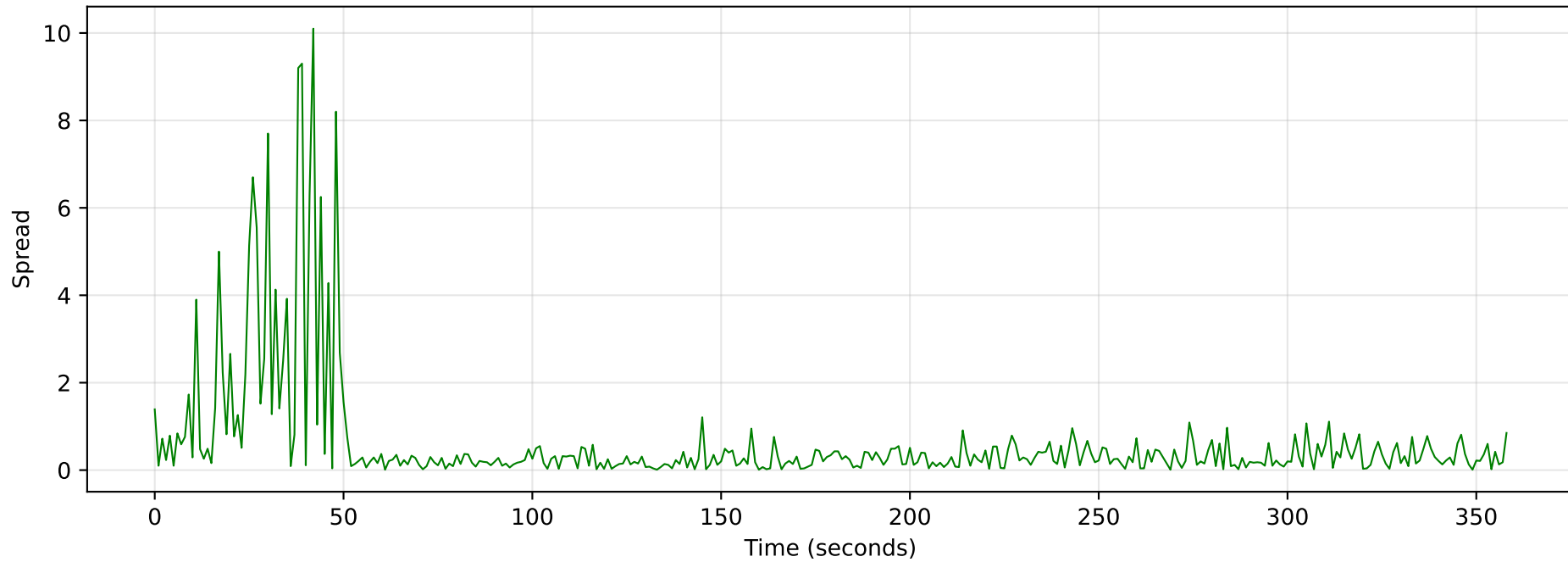


Scenario C: 80 Noise + 20 Mom

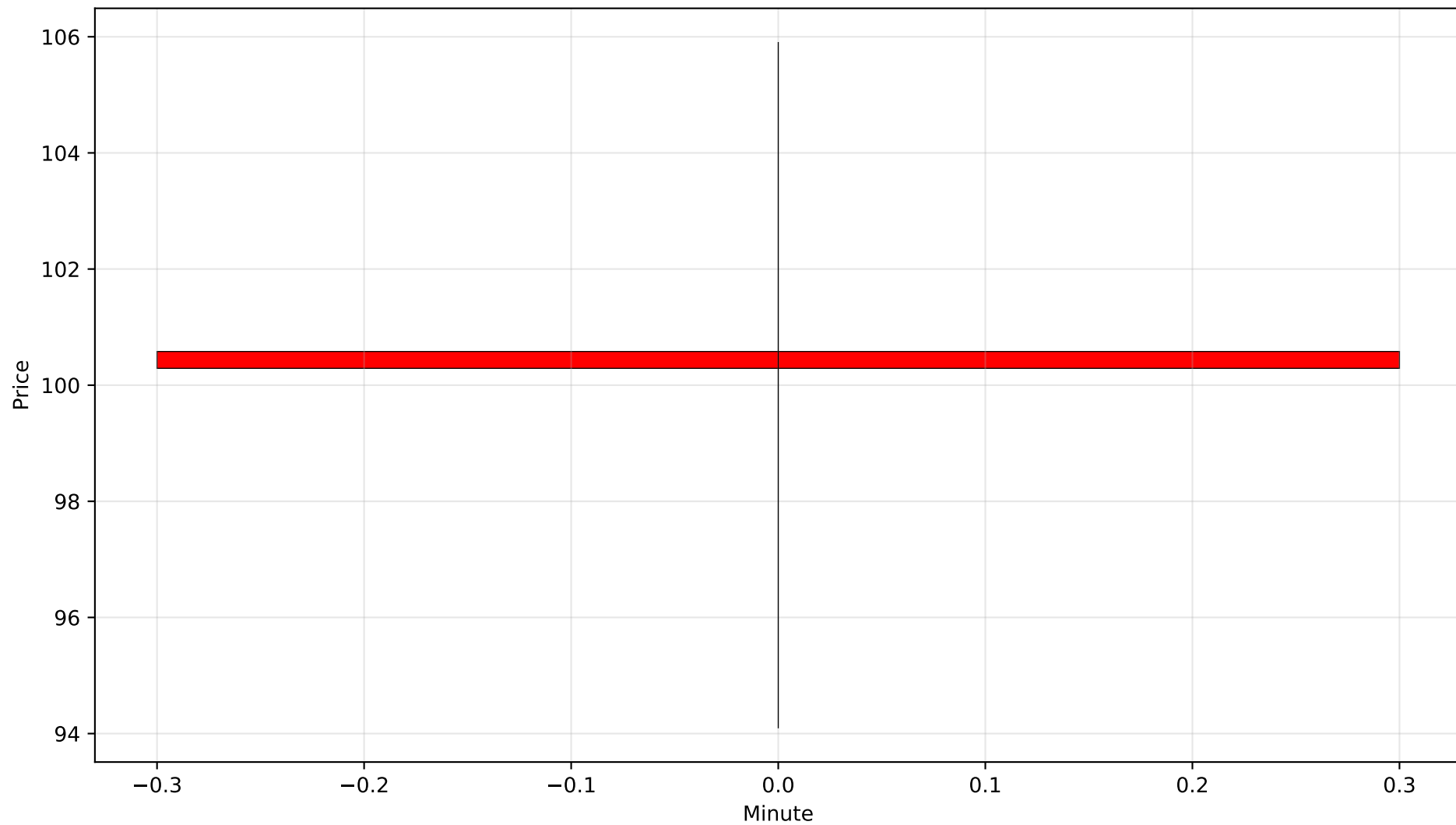
Mid-Price Over Time



Bid-Ask Spread Over Time



Scenario C: 80 Noise + 20 Mom: 1-Min OHLC



Scenario Comparison

Metric	Scenario A (100 Noise)	Scenario B (80 Noise + 20 MM)	Scenario C (80 Noise + 20 Mom)
Total Trades	30347	44821	24552
Total Volume	57216	96485	46247
VWAP	100.0095	77.9735	100.0076
Avg Spread	0.4407	0.4007	0.6082
Volatility	0.0107	0.0055	0.0080

INTERPRETATION OF RESULTS

Scenario A (100 NoiseAgents):

- Pure random order flow creates baseline market dynamics
- Spread fluctuates randomly without systematic tightening
- Volatility reflects Brownian fair value randomness

Scenario B (80 NoiseAgents + 20 MarketMakerAgents):

- Market makers provide two-sided quotes, tightening spreads
- Expected: Lower average spread compared to Scenario A
- Market makers absorb noise trader flow, stabilizing prices

Scenario C (80 NoiseAgents + 20 MomentumAgents):

- Momentum agents amplify trends via positive feedback
- Expected: Higher volatility compared to Scenario A
- Trend-following can create larger price swings

KEY OBSERVATIONS:

- Market makers reduce spread and may reduce volatility
- Momentum traders increase volatility and trend persistence
- Agent composition significantly affects market microstructure

This completes Week 2: Limit Order Book Simulator